

Using orientation tests to solve basic problems on polygons

Vera Sacristán
Rodrigo Silveira

Computational Geometry
Facultat d'Informàtica de Barcelona
Universitat Politècnica de Catalunya

USING ORIENTATION TESTS ON POLYGONS

Point in polygon test

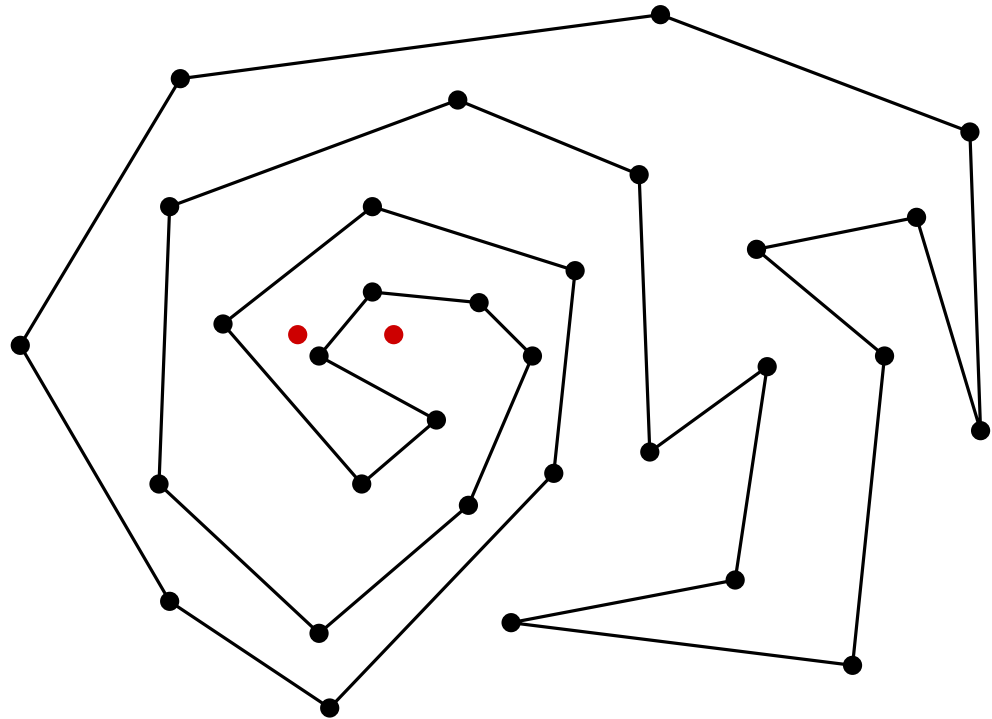
Input:

A polygon $p_1, p_2, \dots, p_n$

A query point q

Output:

Yes/No $q \in P$



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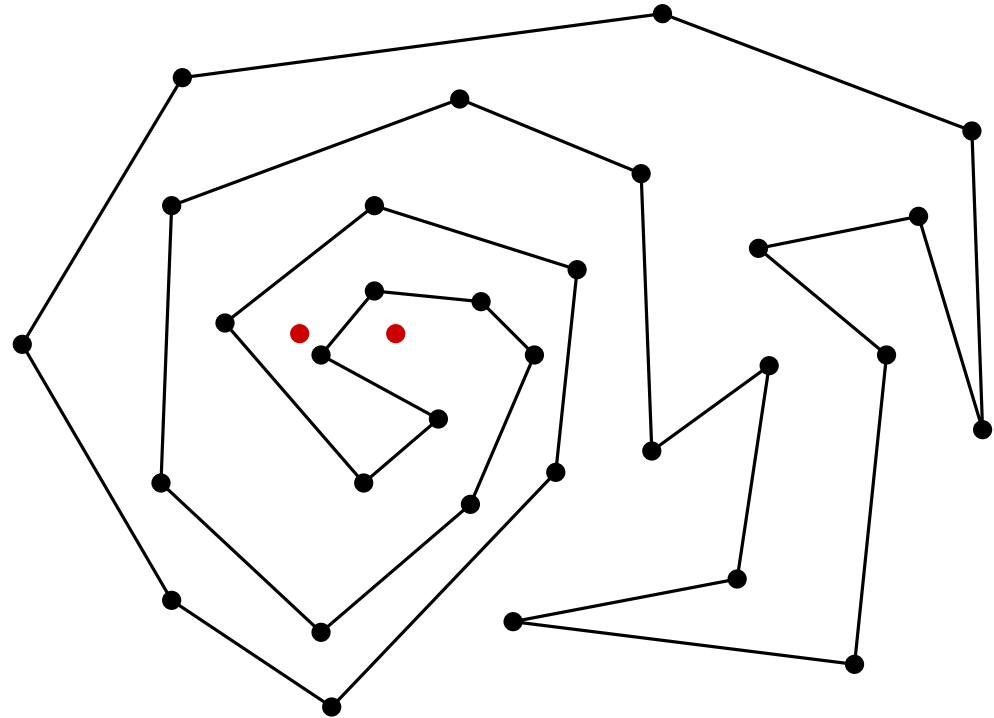
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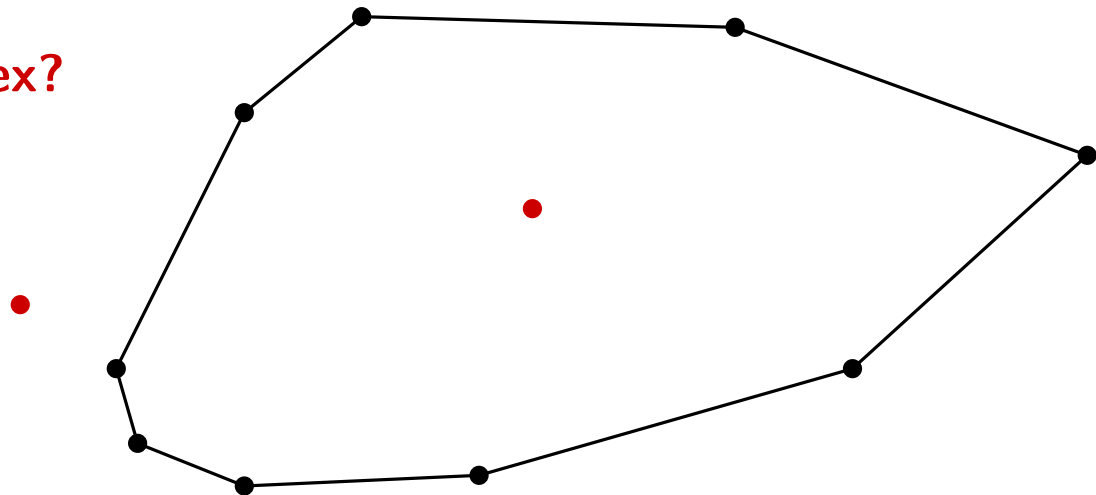
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What if the polygon is convex?



USING ORIENTATION TESTS ON POLYGONS

Intersection test line - polygon

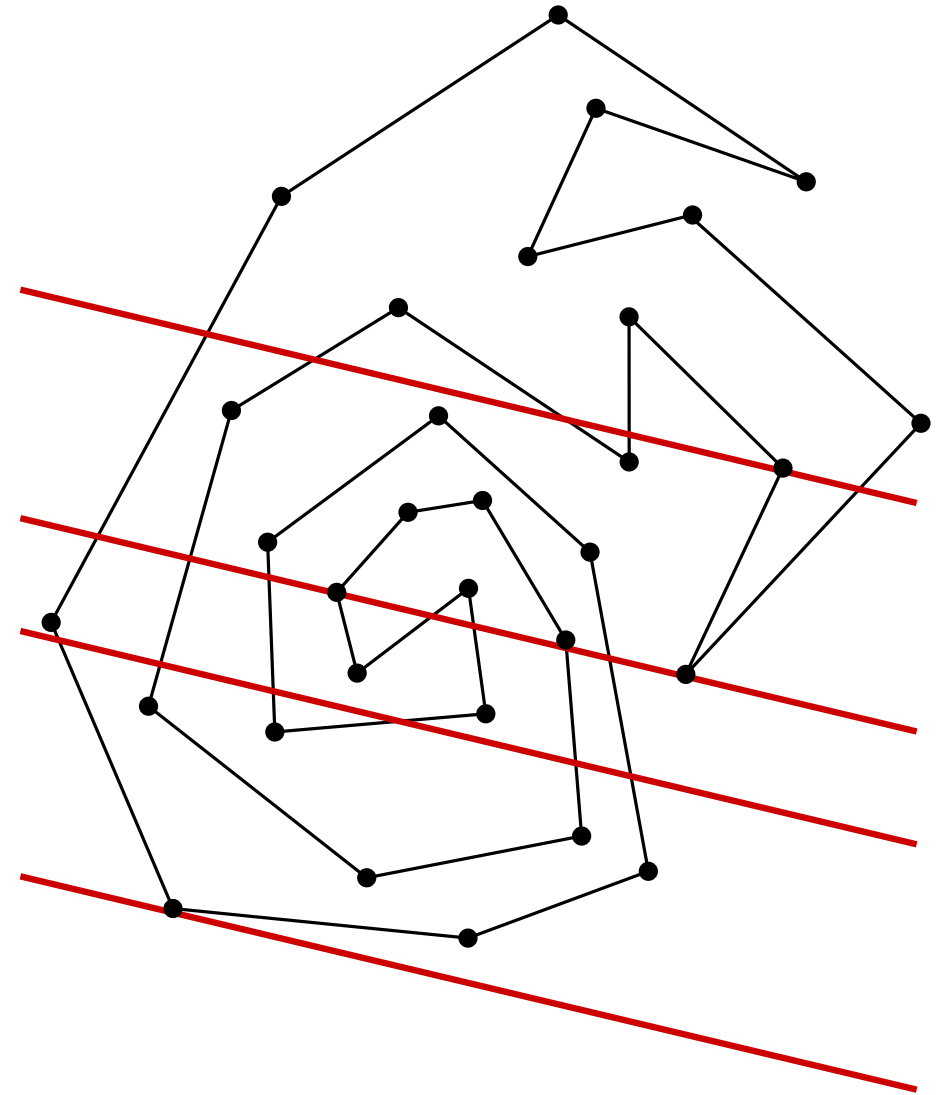
Input:

ℓ : a line (through p and q)

P : a polygon (with vertices $p_1, p_2, \dots, p_m$)

Yes/No they intersect.

If they do, the edges of P intersecting ℓ



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Intersection test line - polygon

Input:

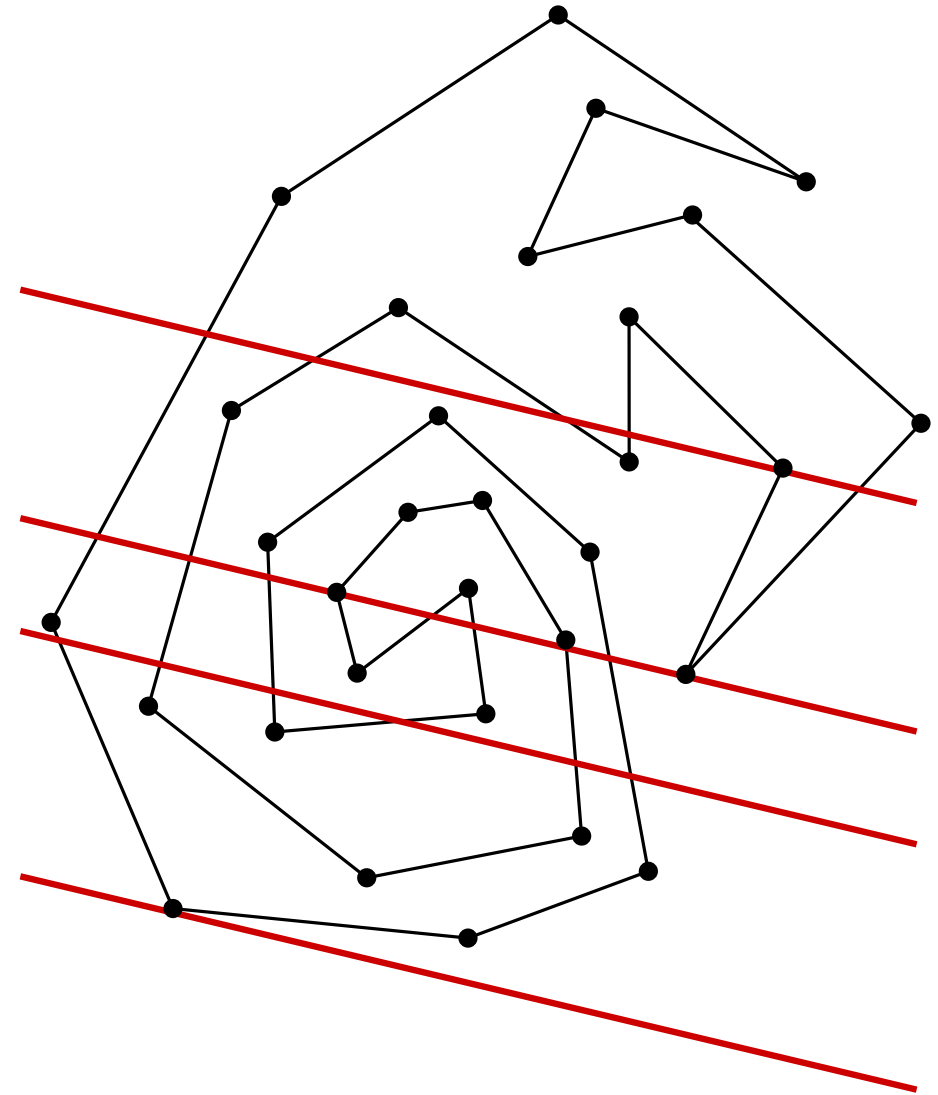
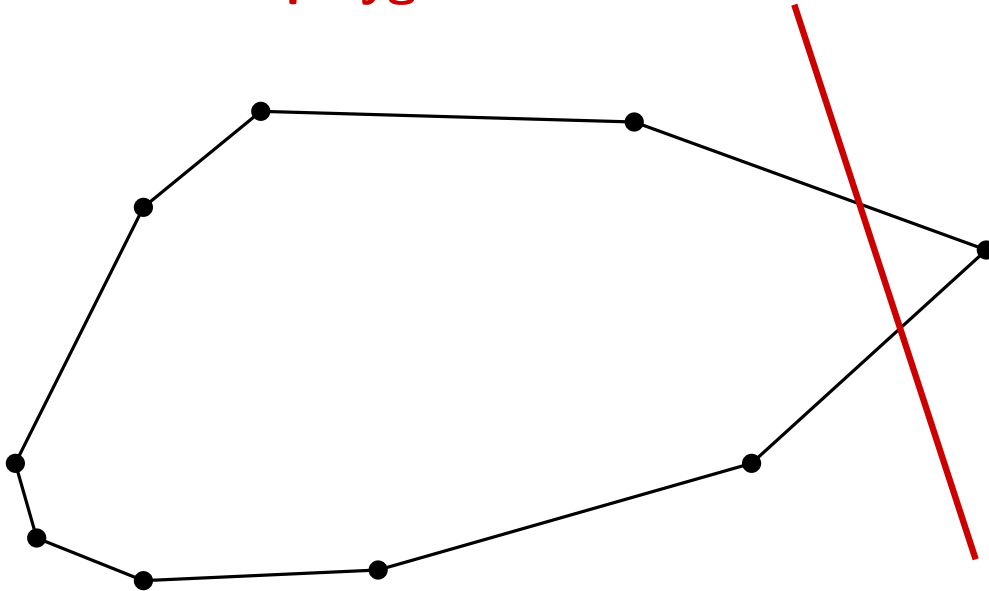
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Supporting lines point - polygon

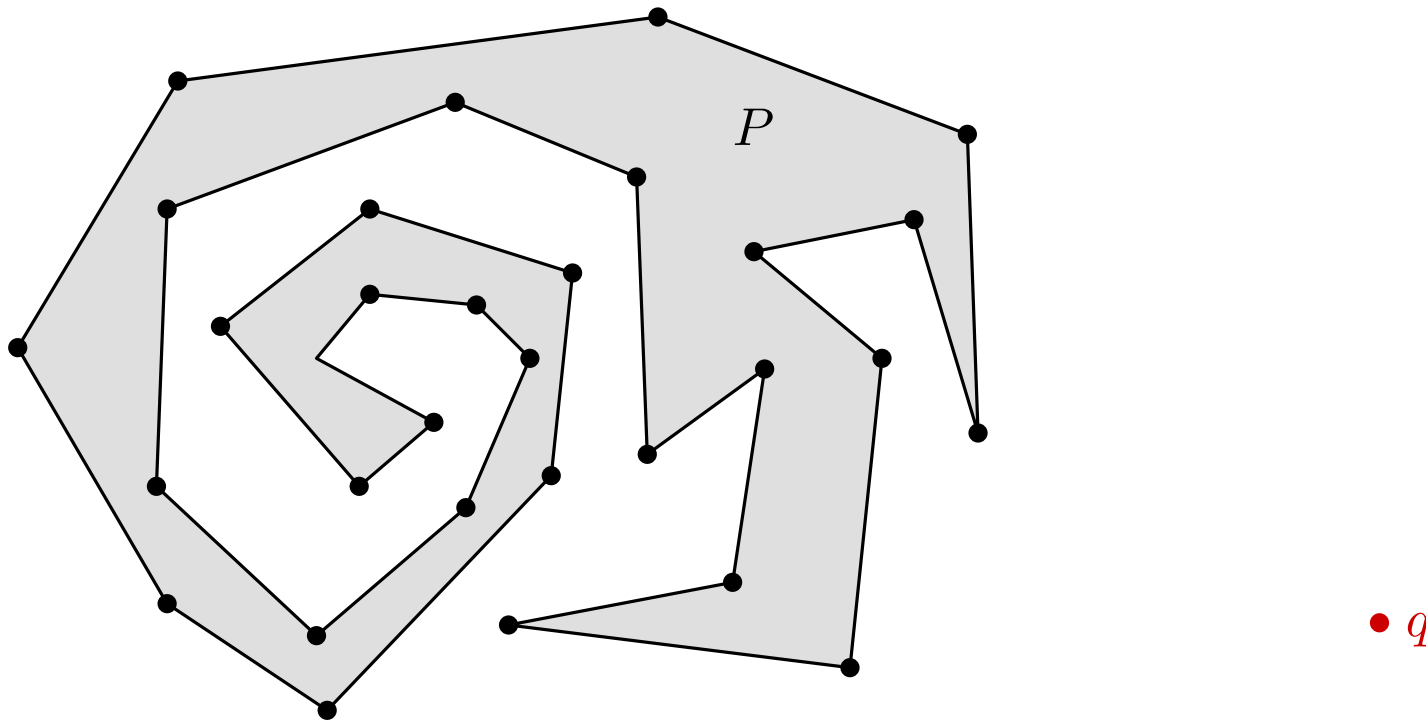
Input:

A polygon P with vertices $p_1, p_2, \dots, p_n$

A point q not belonging to the convex hull of P

Output:

Lines through q and P that leave all of P to one side



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Supporting lines point - polygon

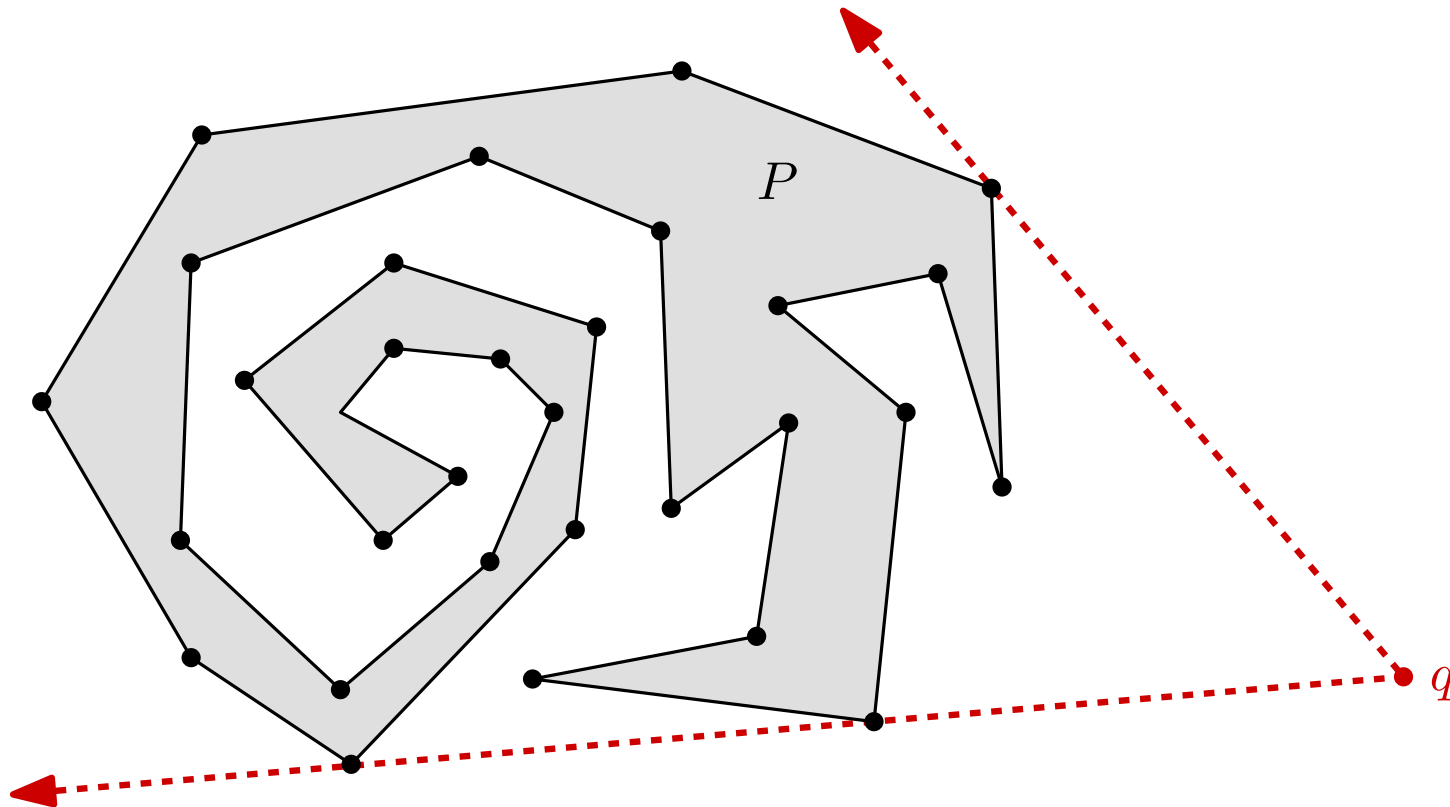
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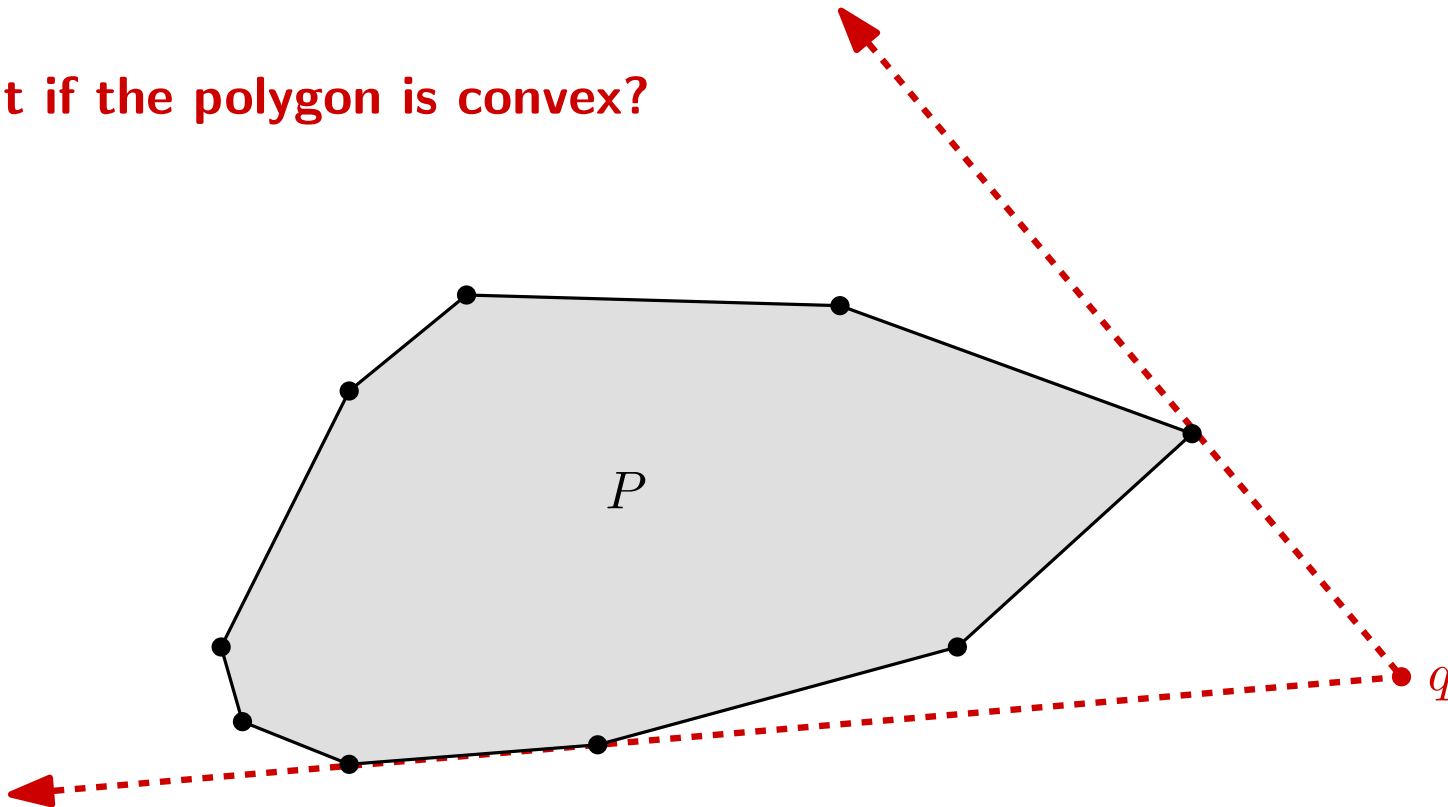
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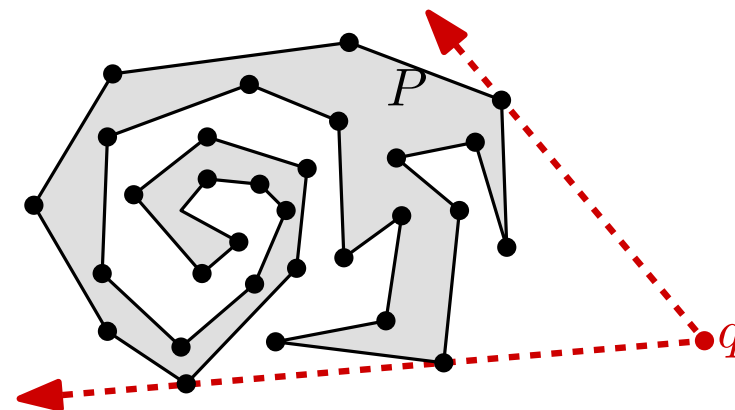
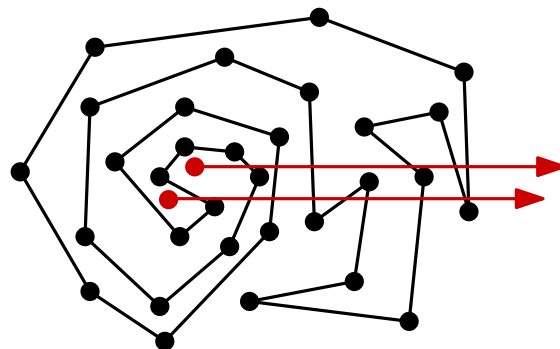
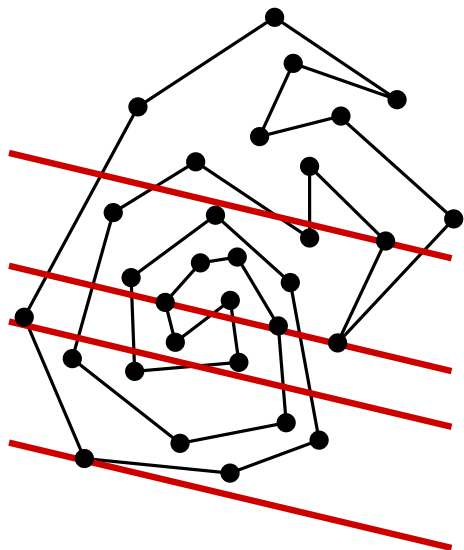


USING ORIENTATION TESTS ON POLYGONS

Key ideas of the solutions – general simple polygons

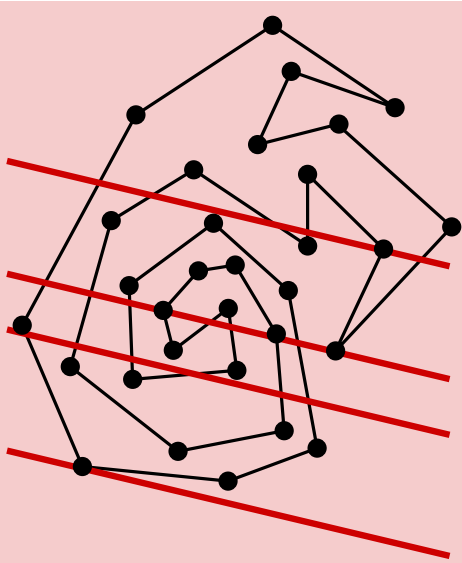
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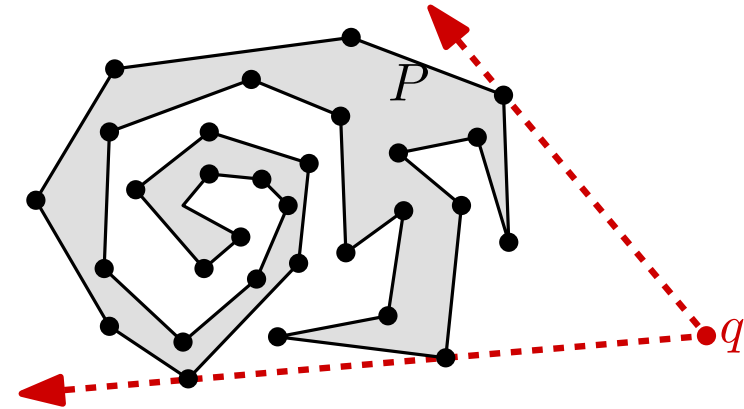
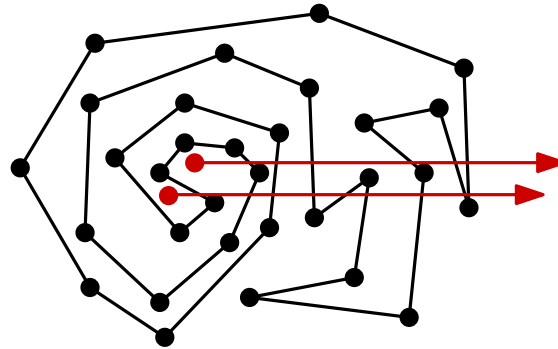


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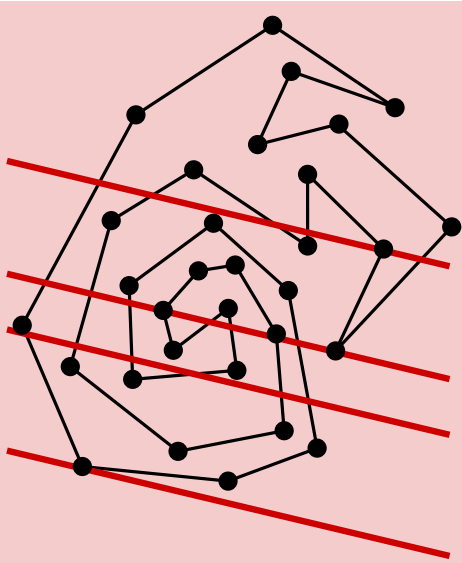


Geometric property:
No particular one.



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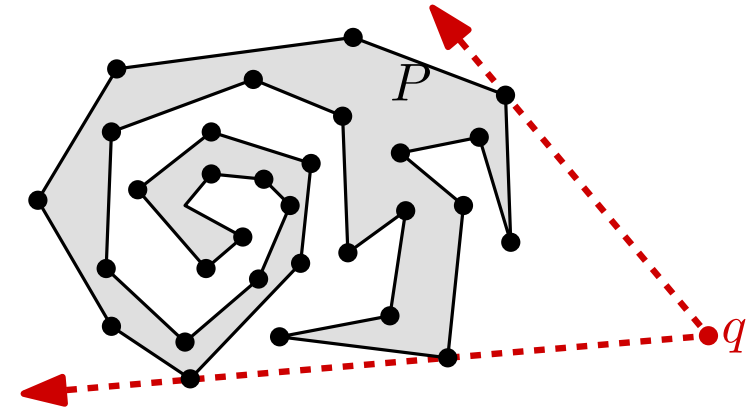
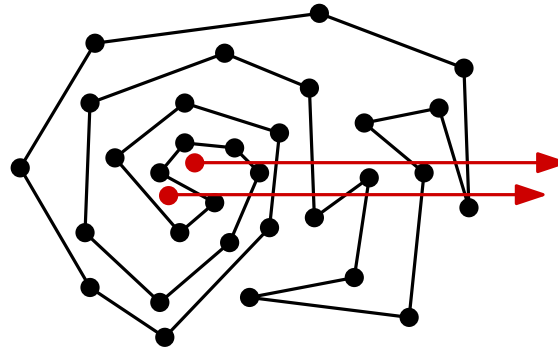
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Brute-force solution

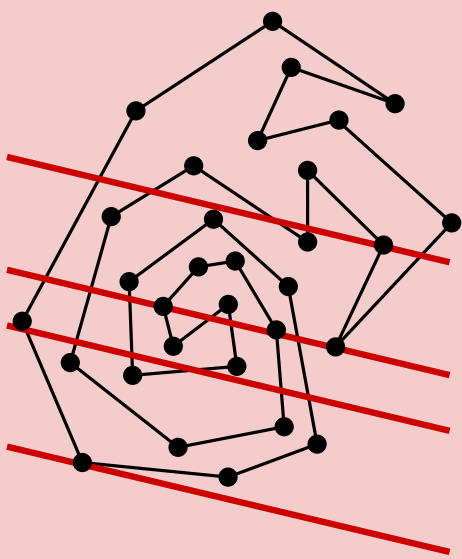
$O(n)$ time

$O(n)$ space



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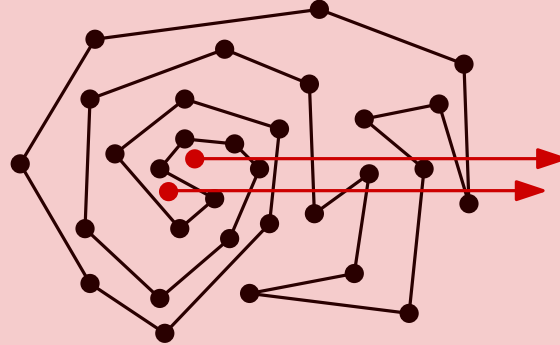


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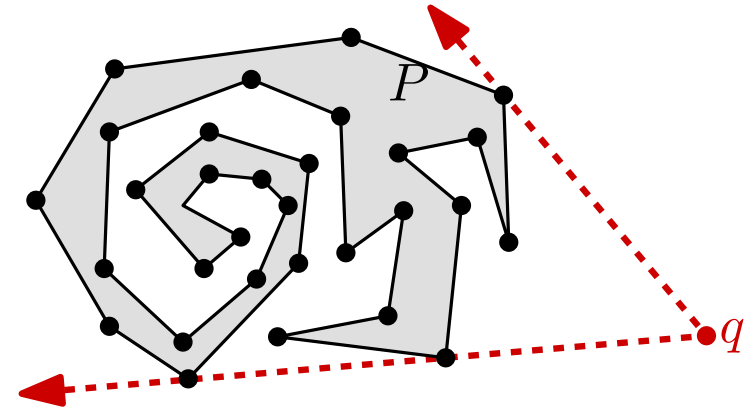


Brute-force solution

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 $O(n)$ space

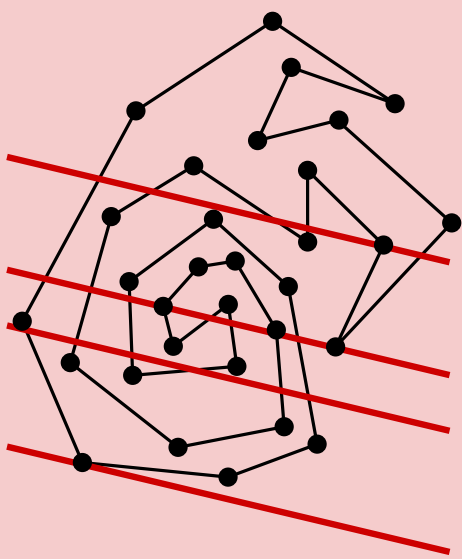


Geometric property:
 $p \in P \Leftrightarrow$ The number of intersections of ∂P and any halfline with origin at p is odd.



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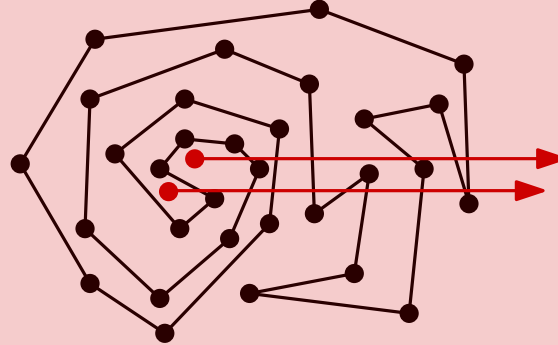


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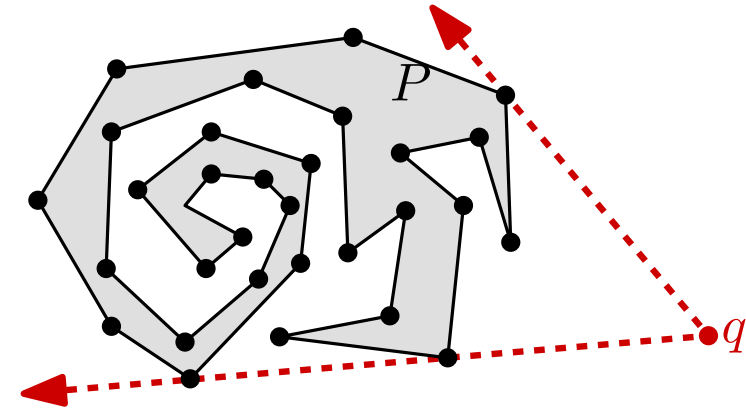


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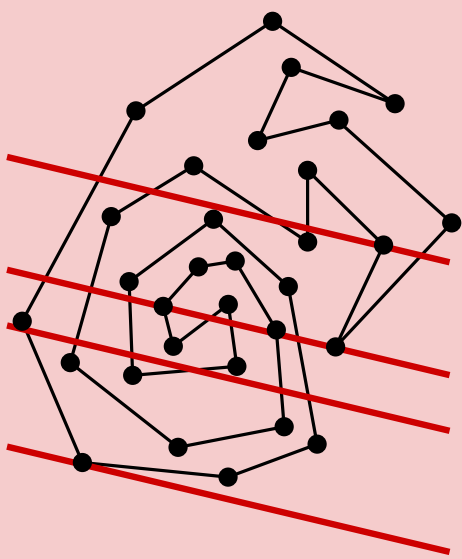
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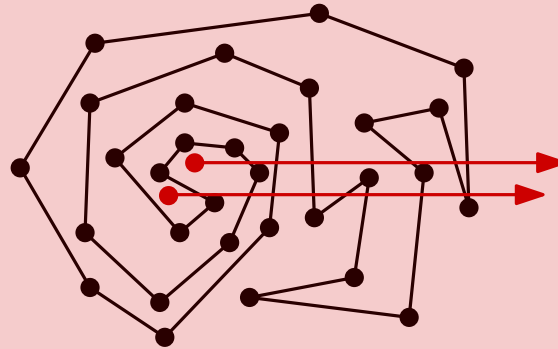


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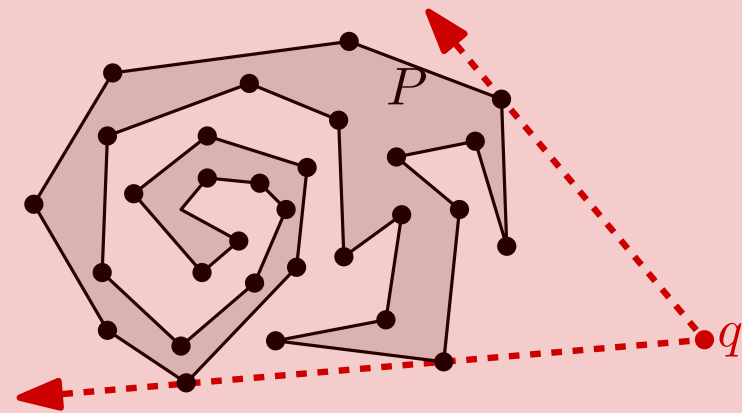


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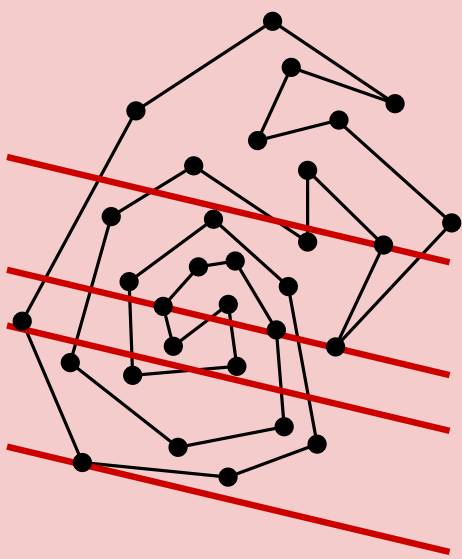
$O(n)$ time
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Geometric property:
The solutions are the angularly extreme vertices of P as seen from p .

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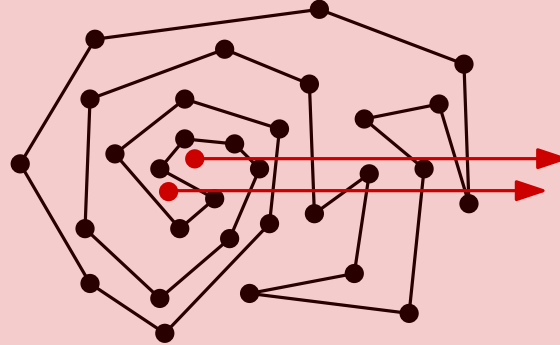


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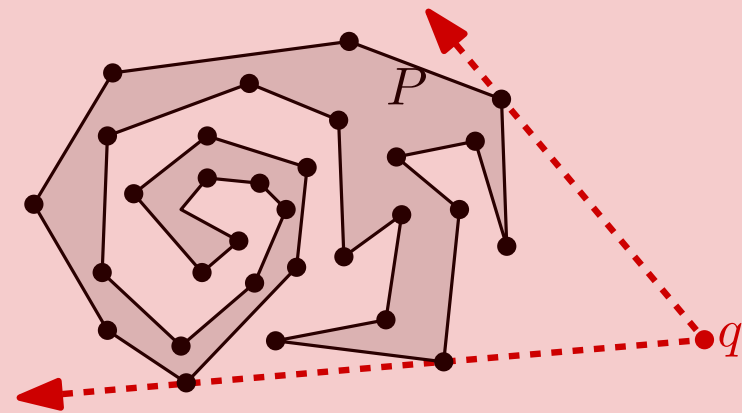


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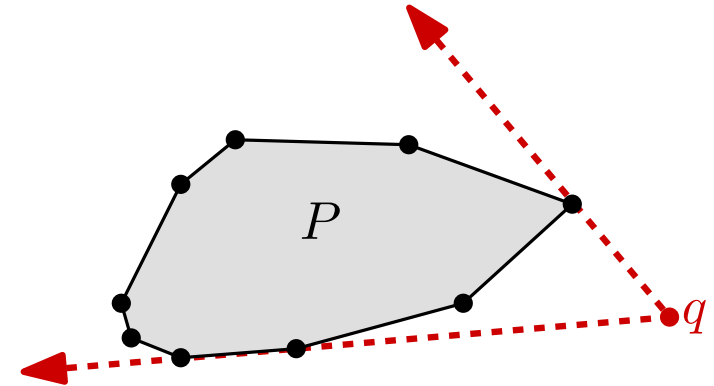
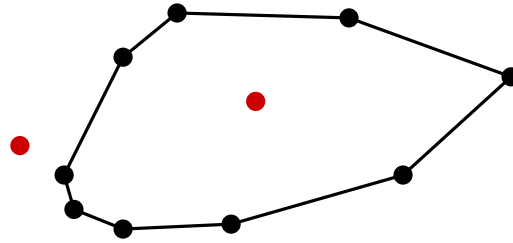
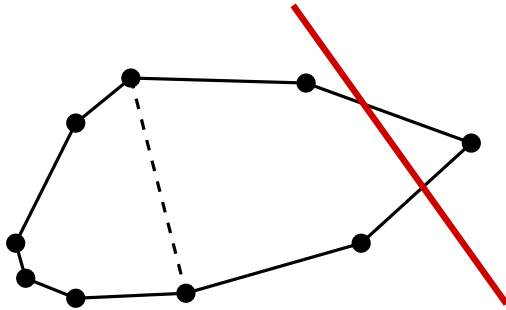


Use a max/min algorithm

$O(n)$ time
 $O(n)$ space

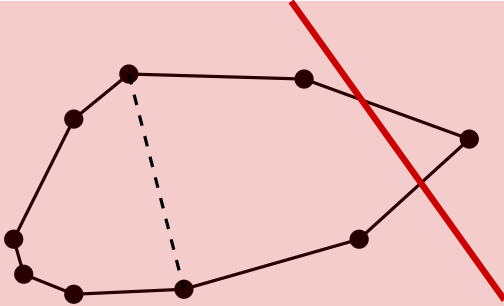
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Key ideas of the solutions – convex polygons

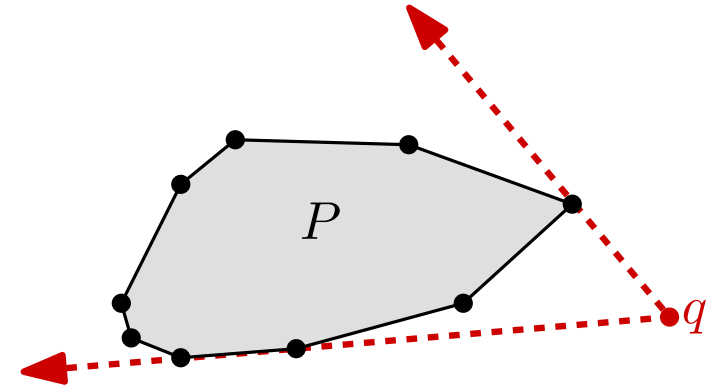
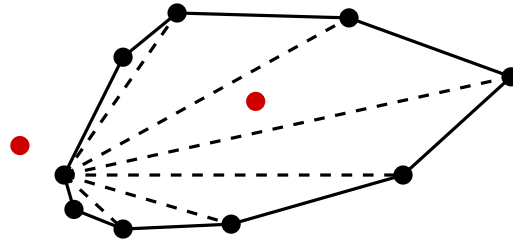


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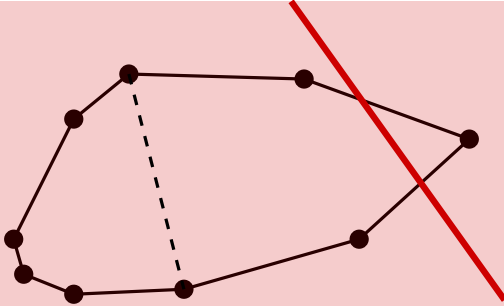


Geometric property:
The “middle segment”
allows us to recurse on
only half of the
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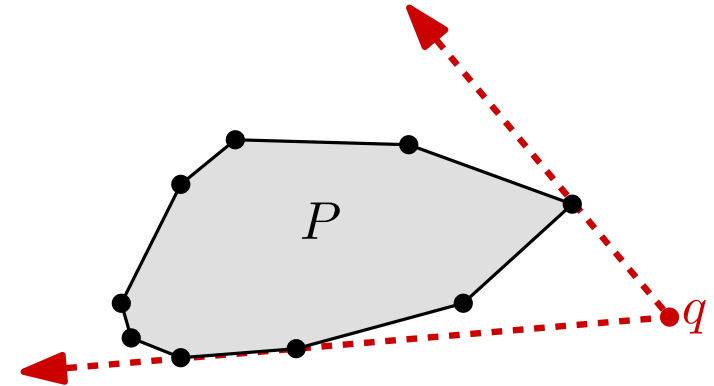
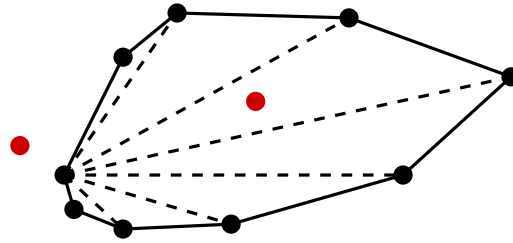


Binary search solution

$O(\log n)$ time

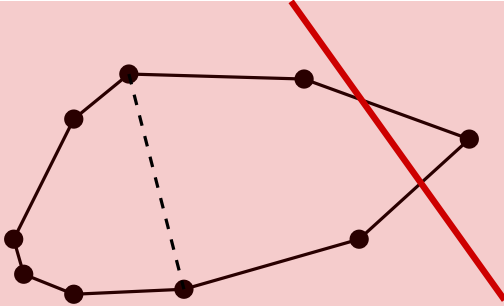
$O(n)$ space

(after preprocess)



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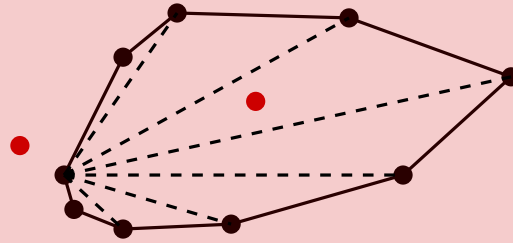


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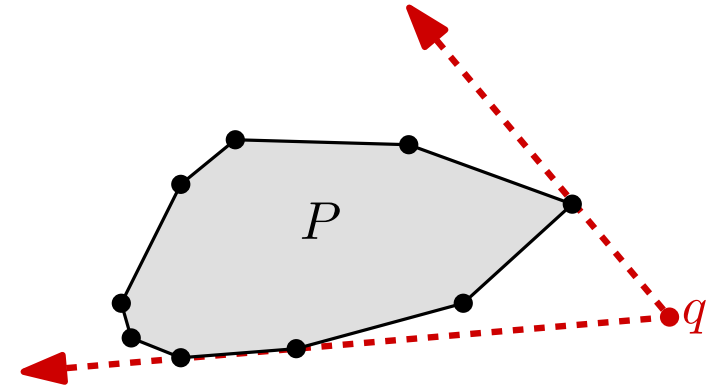
$O(n)$ space

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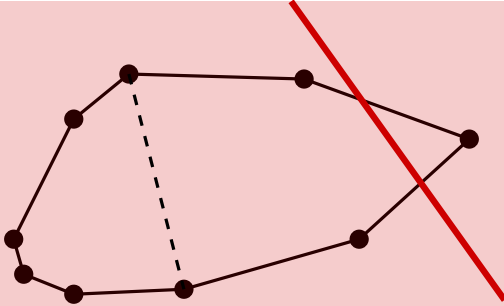
Geometric property:

We can decompose interior of polygon into sorted triangles, and binary search there



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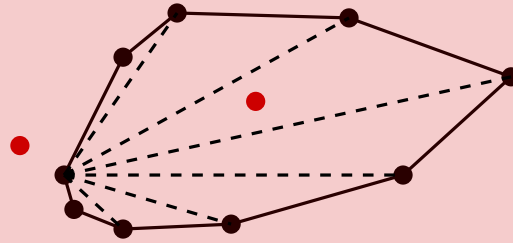


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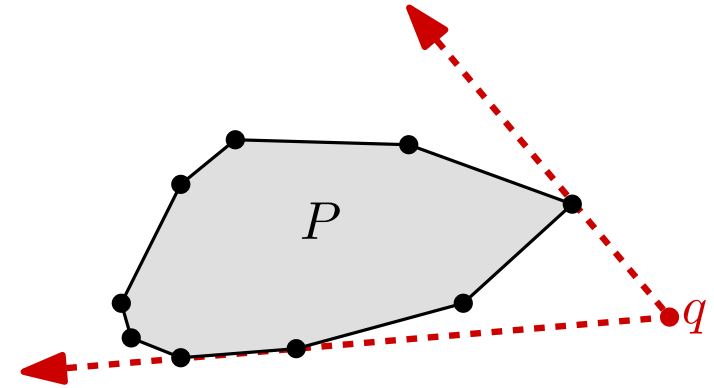


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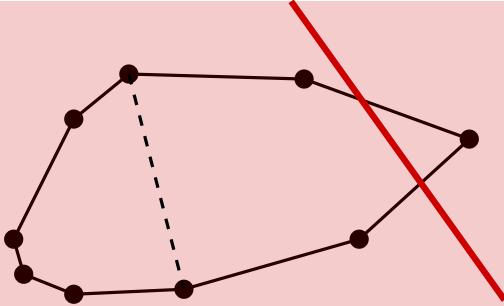
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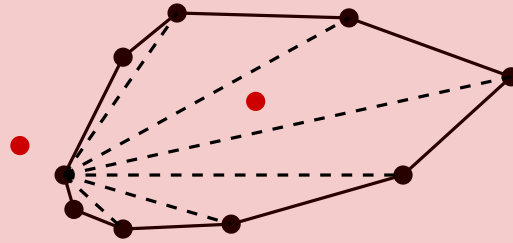


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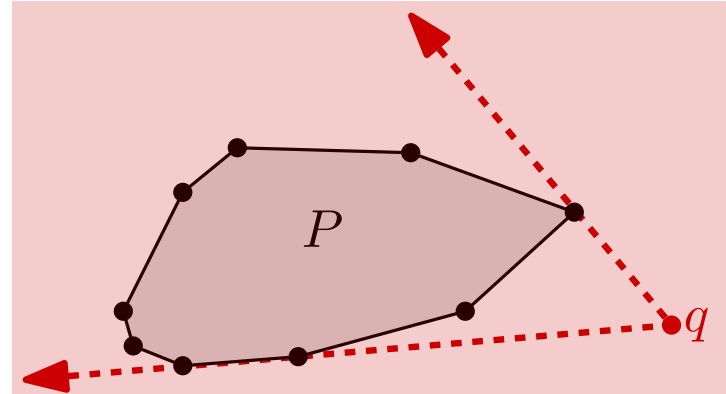


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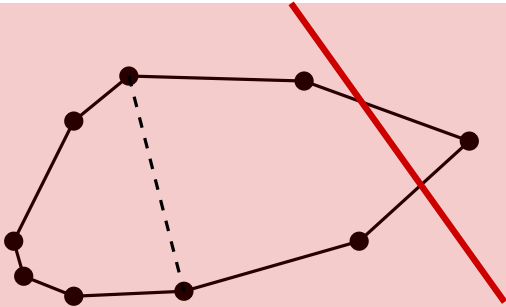


Geometric property:

Angle wrt q is unimodal along ∂P .

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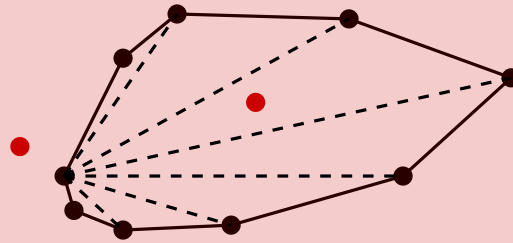


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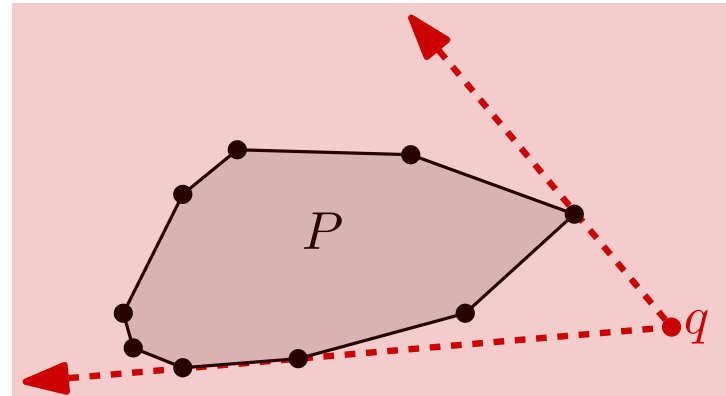


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FURTHER READING

J. O'Rourke

Computational Geometry in C

Cambridge University Press, 1994 (2nd ed. 1998), pp. 17-35.

F. P. Preparata and M. I. Shamos

Computational Geometry: An Introduction

Springer-Verlag, 1985, pp. 36-45.